Inverted MAML Card 1: Round 3 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.02, L2=2560, Mom=0.8, Clamp=±70 LSB

1. Round 3 Summary & Cumulative Comparison (R1–R3)

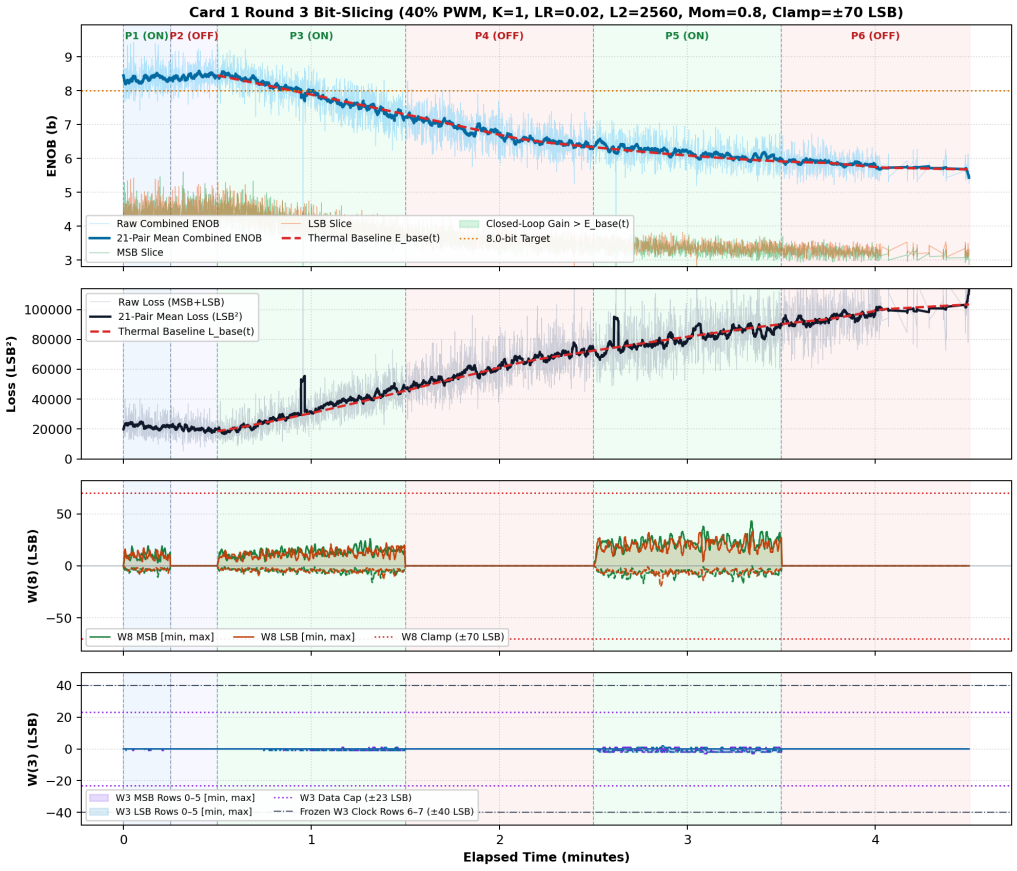
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 3 Detrended Trend Metrics:** Lift vs E_base = **+0.04 b**, Loss Ratio = **1.00x**, Climb = **+0.24 b**, Jitter = **0.34 b**, Recovery = **1.00**, Score = **+0.204**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R1dc_fix	0.03	1024	0.9	±75	+1.00b	1.71x	+0.95b	0.40b	0.00	+1.799
R2dc_fix*	0.035	1536	0.85	±75	+0.99b	1.69x	+1.16b	0.39b	0.00	+1.957
R3dc_fix	0.02	2560	0.8	±70	+0.04b	1.00x	+0.24b	0.34b	1.00	+0.204

2. Phase-by-Phase Telemetry (3,736 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	321	8.25->8.38 (8.35b)	-	22628->21169	[19,17]	1	320
P2	Cold / Regler OFF	292	8.39->8.47 (8.43b)	-	21156->19041	[0,0]	0	0
P3	Warm / Regler ON	1031	8.35->7.47 (7.91b)	-0.05 b	21006->42181	[26,22]	1	1031
P4	Warm / Regler OFF	866	7.24->6.46 (6.74b)	+0.02 b	47268->69834	[0,0]	0	0
P5	Warm / Regler ON	781	6.50->6.04 (6.17b)	+0.08 b	68926->85993	[43,33]	3	781
P6	Warm / Regler OFF	445	5.86->5.76 (5.84b)	+0.02 b	94707->98681	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 3

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±70 LSB, while W(3) rows 0–5 stay within ±23 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 3 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

